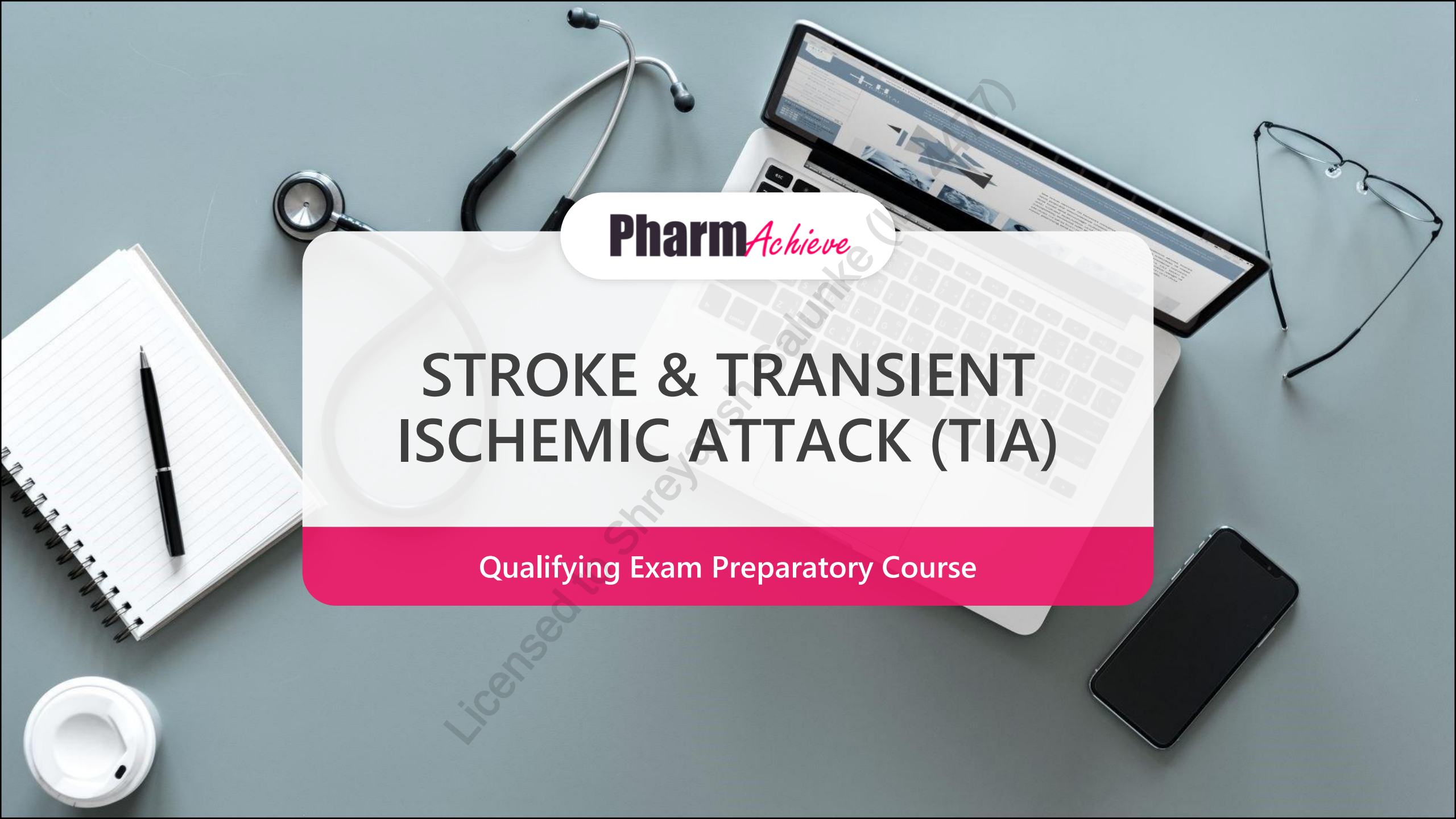




**Pharm***Achieve*

# STROKE & TRANSIENT ISCHEMIC ATTACK (TIA)

Qualifying Exam Preparatory Course

# COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:  
Clinical Care**

**Providing Care:  
Distribution**

**Knowledge  
& Expertise**

**Communication  
& Collaboration**

**Leadership &  
Stewardship**

**Professionalism**

# LEGEND

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**ACEi** Angiotensin Converting Enzyme Inhibitor

**ADR** Adverse Drug Reaction

**ASA** Acetylsalicylic Acid

**BP** Blood Pressure

**CBC** Complete Blood Count

**CT** Computed Tomography

**CVA** Cerebrovascular Accident

**DAPT** Dual Antiplatelet Therapy

**DOAC** Direct Oral Anticoagulant

**ECG** Electrocardiogram

**HRT** Hormone Replacement Therapy

**HTN** Hypertension

**ICH** Intracranial Hemorrhage

**INR** International Normalized Ratio

**LDL** Low Density Lipoprotein

**MOA** Mechanism of Action

**MRI** Magnetic Resonance Imaging

**SBP** Systolic Blood Pressure

**TIA** Transient Ischemic Attack

**TPN** Total Parenteral Nutrition

# DEFINITIONS

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|                                 |                                                                                                                                                                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transient Ischemic Attack (TIA) | <ul style="list-style-type: none"><li>• Disruption in blood flow to the brain resulting in damage</li><li>• Neurological deficit lasting &lt;24 h (usually &lt;30 minutes) and may be reversible</li></ul>                                                                          |
| Stroke                          | <ul style="list-style-type: none"><li>• Same as TIA, except neurological deficit lasting &gt;24 hours and may be irreversible</li></ul>                                                                                                                                             |
| Ischemic Stroke                 | <ul style="list-style-type: none"><li>• Occlusion of blood vessel(s) supplying the brain</li><li>• <u>Penumbra</u>: area of reversible ischemia surrounding an ischemic event that can be saved in the first few hours after an ischemic stroke, if treated appropriately</li></ul> |
| Hemorrhagic Stroke              | <ul style="list-style-type: none"><li>• Bleeding in the brain</li></ul>                                                                                                                                                                                                             |

# PATHOPHYSIOLOGY

Burst  
aneurysm

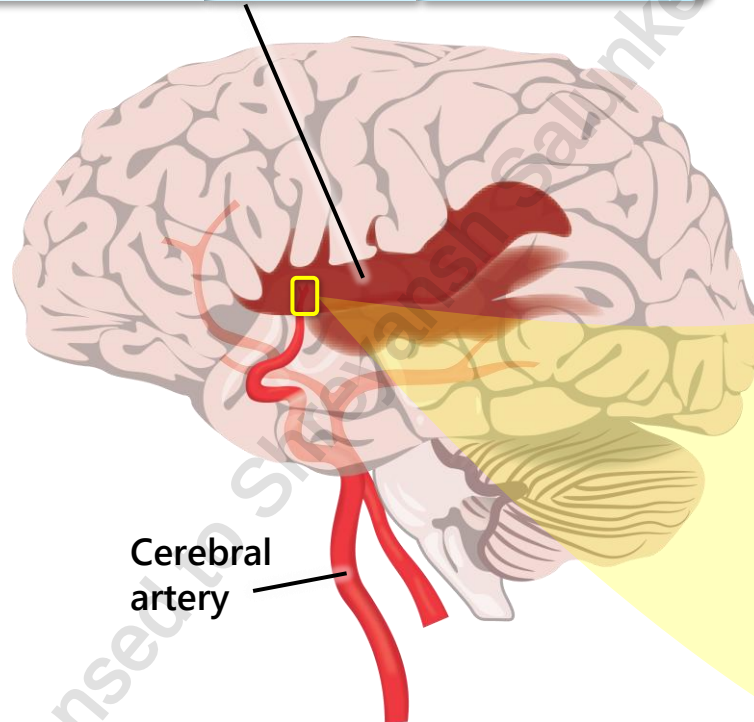
Brain  
bleeding

Excess  
pressure

Brain tissue  
death

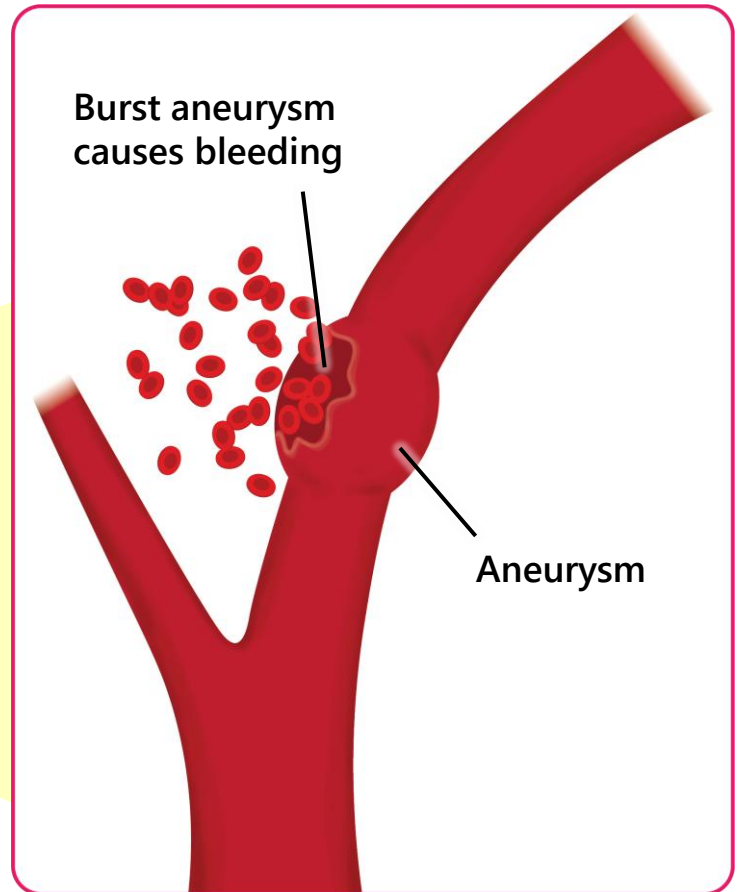
## Hemorrhagic Stroke

Rupture of blood vessel  
causes damage to the  
surrounding tissue  
resulting in an increase  
in intracranial pressure



Burst aneurysm  
causes bleeding

Aneurysm



# PATHOPHYSIOLOGY

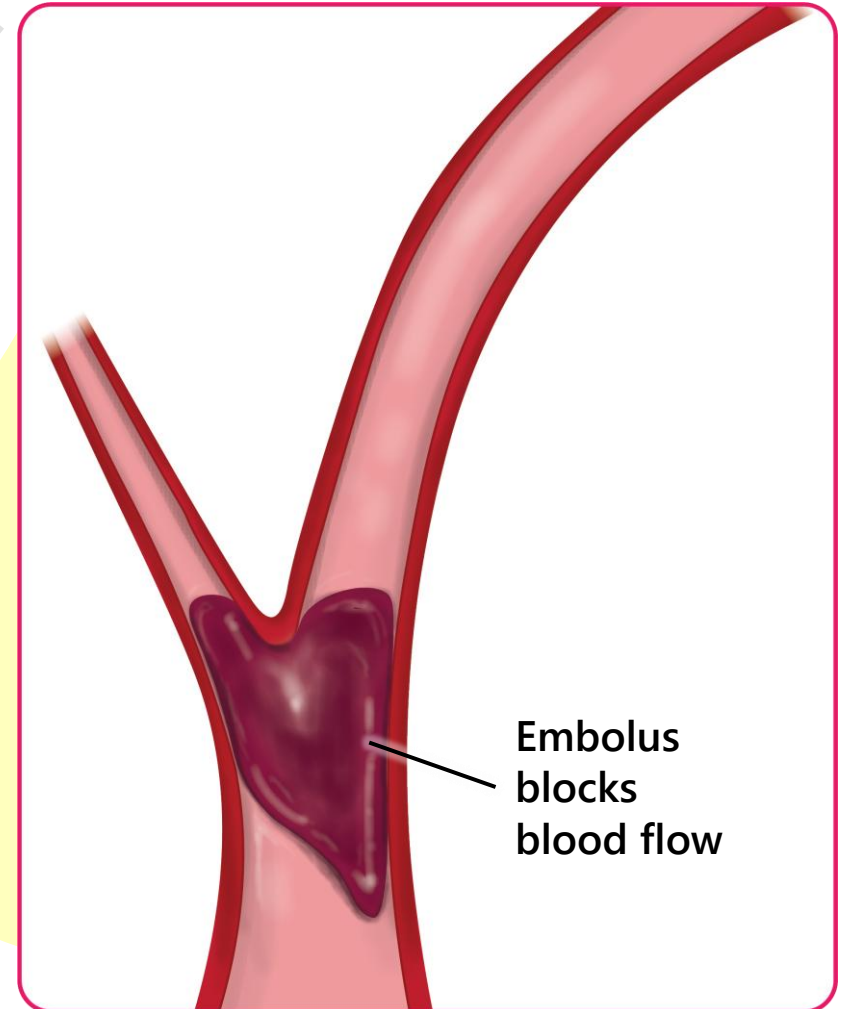
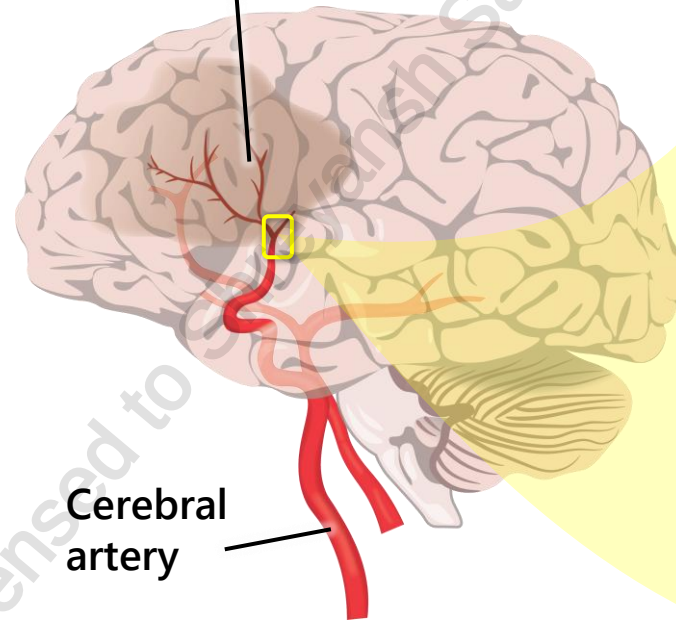
Blood clot breaks away from plaque build-up

Embolus blocks blood flow to brain

Brain tissue death

## Ischemic Stroke

1. **Atherosclerotic origin:** clot formed from ruptured **plaque (cholesterol)** in inner arterial wall
2. **Cardioembolic origin:** clot of **cardiac origin** (e.g. atrial fibrillation) embolizing to the brain



# CLINICAL PRESENTATION

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Sudden onset of:

- Weakness, numbness, and tingling on one side of the body
- Severe and abrupt onset of headache
- Agitation / delirium
- Drooping of the face
- Trouble speaking/slurred speech (more likely if left side of brain is injured)
- Blurred vision, dizziness, and spatial issues (more likely if right side of brain is injured)

**F**

**Face:** is it drooping?

**A**

**Arms:** can you raise both?

**S**

**Speech:** is it slurred or jumbled?

**T**

**Time:** to call 911 right away!



Symptoms depend on location of infarction

# RISK FACTORS

## Hemorrhagic Stroke

- Severe HTN
- Smoking
- Anticoagulants (warfarin)
- Coma
- Cerebral trauma
- Severe headache
- Hyperglycemia ( $>9.4$  mmol/L)

## Ischemic Stroke

### Modifiable

- HTN
- Diabetes
- Low HDL
- Lifestyle
- Smoking
- HRT / oral contraception
- Atrial fibrillation
- Mitral stenosis
- Heart disease

### Non-modifiable

- Age  $>55$  years
- Male  $>$  female (women have  $\uparrow$  mortality)
- Race (African American, Hispanics, Asian, First Nations)
- Family history
- Previous stroke/TIA

# MEDICAL CAUSES

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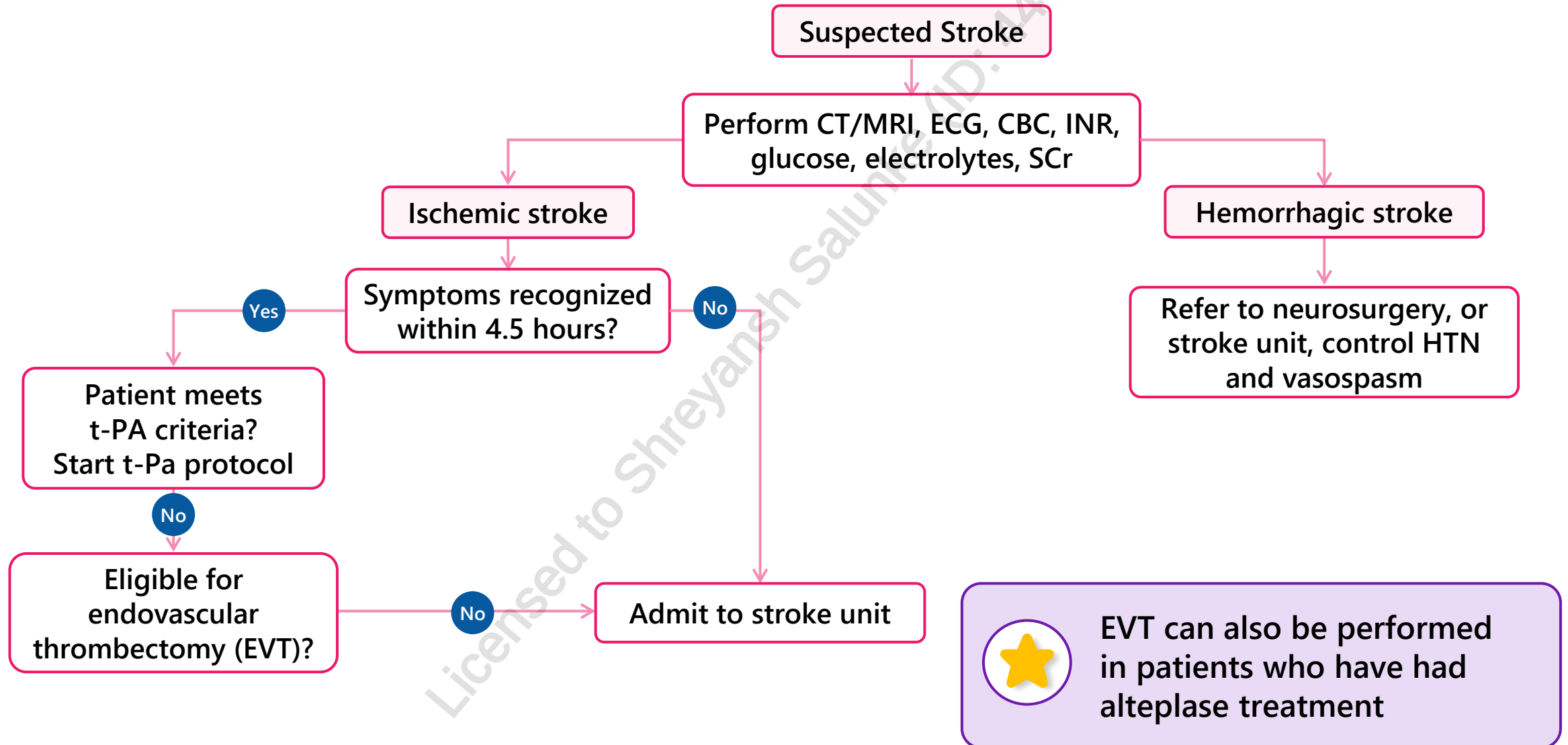
- **Triptans and cocaine:** vasospasm and hypertension
- **Excess alcohol:** presentation may resemble stroke; falls increase chance of hemorrhagic stroke
- **Anticoagulants:** hemorrhagic stroke
  - Warfarin—especially if high INR
  - Novel oral anticoagulants (dabigatran, rivaroxaban)
  - Unfractionated heparin, low molecular weight heparin, fondaparinux
- **Hormone replacement therapy and oral contraceptives:** ischemic stroke

# GOALS OF THERAPY

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- ✓ Treat acute stroke to prevent further debilitation and improve quality of life
- ✓ Prevent recurrent stroke/TIA and associated neurological deficits
- ✓ Reduce risk of stroke/TIA-related mortality
- ✓ Minimize ADR from pharmacological therapy

# TREATMENT ALGORITHM - ACUTE STROKE



# ALTEPLASE (TPA)

## FIRST-LINE THERAPY

MOA: Fibrinolytic agent - converts plasminogen to the proteolytic enzyme plasmin, which lyses fibrin  
→ Intravenous alteplase is cleared primarily by the liver and has an initial half-life of < 5 minutes and a terminal half-life of approximately 72 minutes

Administer within **4.5 hours** of symptom onset → If IV alteplase is considered after 4.5 hours, consult a physician with stroke expertise

- Indicated for patients presenting with ischemic stroke
- Must be 18 years of age or older
- Dosing: 0.9 mg/kg to max of 90 mg infused over 60 minutes
- Administer ASA 160 mg at least 24 hours **after** alteplase once CT excludes intracranial hemorrhage
- No concomitant heparin, warfarin, or aspirin during first 24 hours
- CT/MRI must be completed prior to starting antiplatelet/anticoagulant
- Target door-to-alteplase time is less than 60 minutes
- *Tenecteplase may be considered an alternative agent if within 4.5 hours of stroke symptom onset (not yet Health Canada approved for this indication)*

# CONTRAINDICATIONS TO ALTEPLASE (TPA) THERAPY

## Absolute contraindications (exclusion criteria)

- Any source of active hemorrhage or any condition that could increase the risk of major hemorrhage after alteplase administration
- Any hemorrhage on brain imaging

## Relative contraindications (requiring clinical judgment based upon specific scenario)

### Historical:

- History of ICH
- Stroke or serious head/spinal trauma within last 3 months
- Major surgery (cardiac, thoracic, abdominal or ortho) in the last 2 weeks
- Arterial puncture at non-compressible site in last 7 days

### CT or MRI findings:

- Signs of extensive infarction

### Clinical:

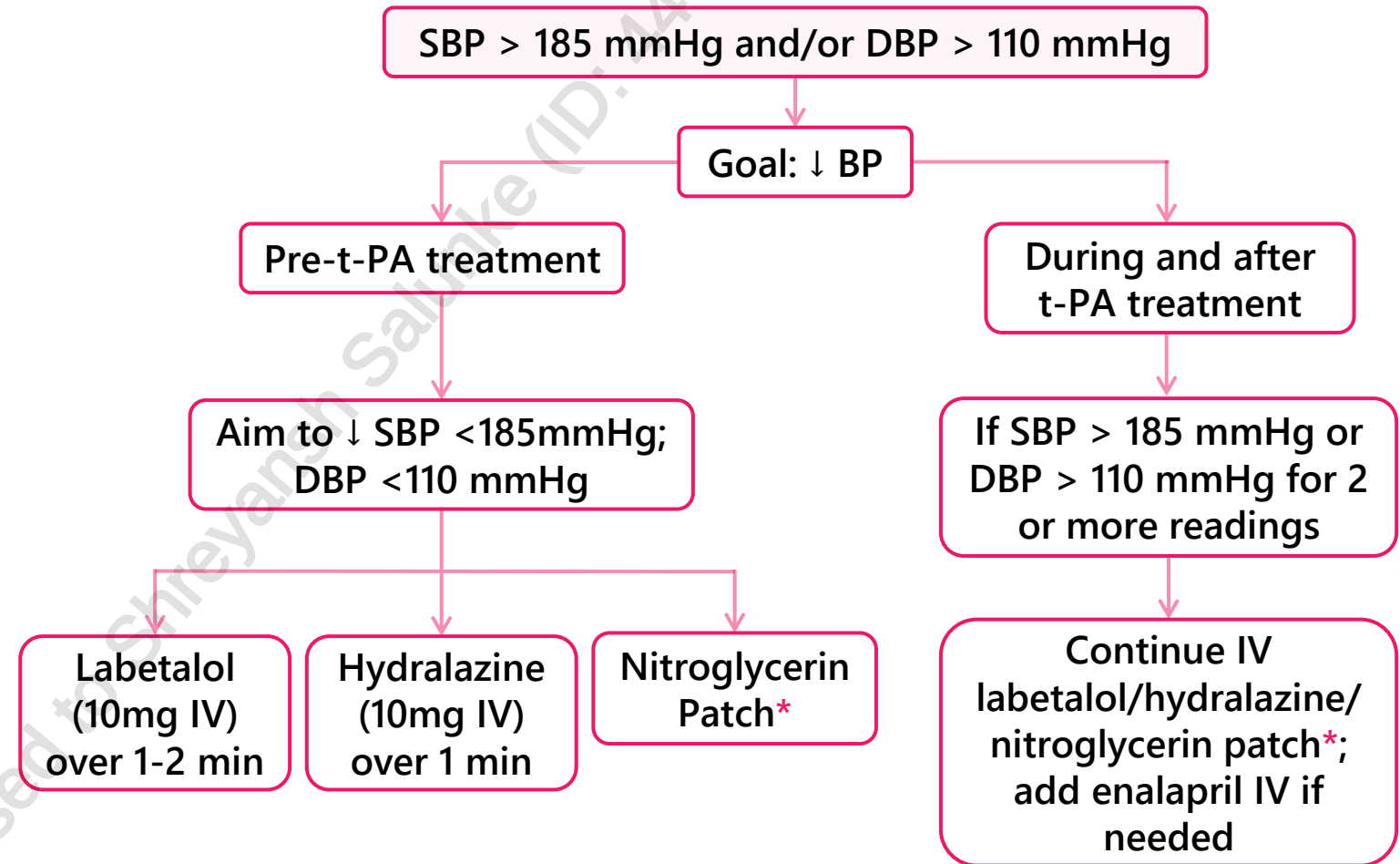
- Symptoms suggestive of subarachnoid hemorrhage
- Currently prescribed and taking a DOAC
- Severe uncontrolled hypertension (SBP > 185 mm HG and/or diastolic BP > 110 mm Hg)

### Laboratory

- BG below 2.7 or above 22.2 mmol/L
- Elevated PTT
- INR greater than 1.7
- Platelet count below 100,000 /mm<sup>3</sup>

# HYPERTENSION MANAGEMENT

- Blood pressure should be lowered and maintained **below 185/110** mmHg before alteplase therapy and kept below 185/110 mmHg for the next 24 hours after administration
- The target level of blood pressure to achieve and maintain in the hyperacute phase of stroke is unknown at this time – choice of antihypertensives and routes of administration should be chosen to **avoid aggressive reduction in blood pressure**



**Do not** aggressively lower BP acutely to target level of 140/90 mmHg due to risk of hypoperfusion

\* Administer within 6 hours of stroke onset

# ACUTE STROKE MANAGEMENT

## ACUTE ANTIPLATELET THERAPY

- Patients **not already** on **antiplatelet therapy** pre-stroke **and not going to be receiving alteplase**, ASA 160 mg as initial loading dose once brain imaging (i.e. CT scan) excludes hemorrhage, then low-dose ASA continued long-term for secondary prevention (more to come on this in upcoming slides!)
- In acute high-risk TIA or minor ischemic stroke (NIHSS score 0-3) of non-cardioembolic origin who are not at a high risk of bleeding, DAPT with ASA and clopidogrel may be considered for the first 3-4 weeks, followed by low-dose ASA or clopidogrel monotherapy indefinitely
  - ASA and ticagrelor may be considered as an alternative
- Long-term DAPT may be considered in certain patients, however, it is generally not recommended for long-term secondary prevention due to bleeding risk
- In patients receiving alteplase, must wait until after the 24-hour post-thrombolysis CT scan excludes intracranial hemorrhage

# STROKE PREVENTION

## Primary/Secondary Prevention

- Lifestyle modification (balanced diet, sodium intake, exercise, weight, and alcohol intake)
- BP – single most important modifiable risk factor!!!
  - <140/90 mmHg
  - <130/80 mmHg (diabetes)
- HbA1C <7% (diabetes)
  - Fasting BG: 4-7 mmol/L
  - 2 h post-prandial BG: 5-10mmol/L

## Secondary Prevention

- Lipid management with statin therapy in ischemic stroke or TIA patients; target LDL <1.8 mmol/L, consider adding ezetimibe +/- PCSK9 inhibitor to reach target
- Initiate ACEi + thiazide diuretic once the patient is stable post-stroke\* to improve stroke outcomes
- Initiate antiplatelet/anticoagulant therapy 24h after tPA administration once CT excludes hemorrhage
  - Antiplatelets are the drug of choice
    - Exception: Patients with atrial fibrillation or other indications for anticoagulation

\*May apply to patients without hypertension as well, but is less clear in the guidelines

# STROKE PREVENTION

## ANTIPLATELET THERAPY

| Drug                                     | Comments                                                                       |
|------------------------------------------|--------------------------------------------------------------------------------|
| ASA                                      | 75-325 mg has equal efficacy (81 mg po daily most commonly prescribed)         |
| Clopidogrel                              | Concomitant PPI use ↓ efficacy (omeprazole and esomeprazole should be avoided) |
| Aggrenox (Dipyridamole 200 mg/ASA 25 mg) | Cancelled post market in Canada in 2020                                        |

# MANAGING SWALLOWING AND NUTRITION

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Bedside dysphagia assessment should be performed soon after a stroke. This is best performed by an experienced dysphagia team using a valid screening tool, and an individualized plan should be implemented.

Give nothing by mouth if any of the following are present:

- Reduced level of consciousness
- Suspected brainstem stroke
- Severe dysarthria
- Wet voice
- Weak cough
- Impaired palatal sensation
- Inability to sit
- Suspected aspiration

Behavioural interventions (such as oropharyngeal exercises) should be considered

A texture-modified diet and fluid consistency is appropriate for dysphagic patients at lower risk of aspiration. Additional IV fluids are often necessary for these patients

Tube feeding (usually nasogastric tube) - if significant aspiration is demonstrated or suspected

- If a tube cannot be tolerated or cannot be placed, consider short-term partial parenteral nutrition or TPN

If swallowing does not improve and the patient is on prolonged enteral nutrition (i.e.  $\geq 4$  weeks), a percutaneous endoscopic gastrostomy (PEG) should be considered

# MONITORING PARAMETERS - SECONDARY PREVENTION

| Efficacy Endpoints   |                                          |                     |
|----------------------|------------------------------------------|---------------------|
| Parameter            | Degree of Change                         | Timeframe           |
| BP                   | < 140/90 mmHg<br>If DM <130/80 mmHg      | 2-3 weeks           |
| Glucose              | Fasting 4-7, 2h post-prandial 5-10mmol/L | 1-2 weeks           |
| HbA1C                | <7%                                      | 3-6 months          |
| LDL                  | <1.8 mmol/L                              | 6-8 weeks           |
| Recurrent stroke/TIA | Prevented                                | Duration of therapy |

| Safety Endpoints                                             |                                                  |                     |
|--------------------------------------------------------------|--------------------------------------------------|---------------------|
| Parameter                                                    | Degree of Change                                 | Timeframe           |
| Signs and symptoms of antiplatelet/<br>anticoagulant therapy | Prevented                                        | Duration of therapy |
| INR (if on warfarin)                                         | 2-3 (2.5-3.5 if mechanical mitral/aortic valves) | Duration of therapy |

# CASE PRACTICE

---

SJ is a 64-year-old man who frequently visits your community pharmacy. Today, he is presenting to the Emergency Department. SJ's wife noted the acute onset of slurred speech and left-sided weakness beginning approximately 35 minutes ago. She called EMS for transport to the nearest hospital. SJ has a past medical history of dyslipidemia, gout, and diabetes for which he takes rosuvastatin 10 mg daily, metformin 1000 mg BID, and allopurinol 200 mg daily. SJ has no allergies to medications. His social history reveals that he is a retired office worker and a former smoker who quit 10 years ago. He consumes approximately 3-4 glasses of wine per week, has an irregular exercise routine, and consumes a diet high in red meat and processed foods.

1. What is the first step in caring for SJ?

- a) ASA 325 mg PO STAT
- b) Enoxaparin 40 mg SC STAT
- c) tPA 0.9 mg/kg
- d) Perform a CT scan of the brain



# CASE PRACTICE

---

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2. All of the following are absolute or relative contraindications to tPA *except*:

- a) Thrombocytopenia
- b) INR 1.6 secondary to warfarin
- c) BP of 200/120 mmHg
- d) Blood glucose of 24.3 mmol/L



# CASE PRACTICE

---

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3. What therapy should be initiated prior to SJ leaving hospital (assuming a diagnosis of ischemic, non-embolic stroke and elevated blood pressure)?

- a) Metoprolol
- b) Perindopril + Indapamide
- c) Amlodipine
- d) Dabigatran



# CASE PRACTICE

---

SJ is a 64-year-old man who frequently visits your community pharmacy. Today, he is presenting to the Emergency Department. SJ's wife noted the acute onset of slurred speech and left-sided weakness beginning approximately 35 minutes ago. She called EMS for transport to the nearest hospital. SJ has a past medical history of dyslipidemia, gout, and diabetes for which he takes rosuvastatin 10 mg daily, metformin 1000 mg BID, and allopurinol 200 mg daily. SJ has no allergies to medications. His social history reveals that he is a retired office worker and a former smoker who quit 10 years ago. He consumes approximately 3-4 glasses of wine per week, has an irregular exercise routine, and consumes a diet high in red meat and processed foods.

4. Which of the following is not a risk factor for ischemic stroke?

- a) Diabetes
- b) Age
- c) Hypertension
- d) Trauma



# REFERENCES

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1. Demaerschalk B, Kleindorfer D, Adeoye O et al. Scientific Rationale for the Inclusion and Exclusion Criteria for Intravenous Alteplase in Acute Ischemic Stroke. *Stroke*. 2016;47:00-00. DOI: 10.1161. <http://stroke.ahajournals.org/content/strokeaha/early/2015/12/22/STR.0000000000000086.full.pdf>.
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3. Heran M, Shamy M, Lindsay P. Canadian Stroke Best Practice Recommendations Acute Stroke Management. 7<sup>th</sup> ED. 2022. Heart and Stroke Foundation Canada. <https://heartstrokeprod.azureedge.net/-/media/1-stroke-best-practices/acute-stroke-management/csbpr7-acute-stroke-management-module-final-eng-2022.ashx?rev=f9b0894446bc4a41a405ae47a62d34ce>
4. Goupil R, Tsuyuki RT, Santesso N, et al. Hypertension Canada guideline for the diagnosis and treatment of hypertension in adults in primary care. Toronto, ON: Hypertension Canada; May 2025. <https://hypertension.ca/wp-content/uploads/2025/05/Hypertension-guideline.pdf>
5. Bath PMW, Song L, Silva GS, et al. Blood pressure management for ischemic stroke in the first 24 hours. *Stroke*. 2022;53(4):1074-1084. doi:10.1161/STROKEAHA.121.036143.
6. <https://www.ncbi.nlm.nih.gov/books/NBK499977/> - Alteplase mechanism of action – Accessed on Feb 1, 2026

# CHANGE LOG

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## July 2025

- Minor formatting changes and reference updated
- Slide 8: Minor edit to hemorrhagic stroke risk factors
- Slide 15: added "clopidogrel" and "indefinitely" to point #2 per guidelines
- Slide 16: added "+/- PCSK9 inhibitor" to match dyslipidemia guidelines
- Slides 19-22: Edits made to case and questions

## September 2025

- Slide 13: added units to platelet count
- Slide 14: Added note on administration window of nitroglycerin patch

## January/February 2026

- Formatting changes made throughout
- Slide 2: Updated new NAPRA competencies
- Slide 12: Added Mechanism of action
- Slide 18: Added new slide on managing dysphagia